CondorBrain™ Institutional Engineering v2.2 DevTeam Update and Major Repo Revision

Sub-Title: *Continuous-Time Physics, Path-Dependency, & Neural CDE Dynamics for SPY Options*

Executive Abstract: The Paradigm Shift

The deployment of CondorBrain v2.2 marks a fundamental phase transition from discrete-time sequence modeling to **Continuous-Time Neural Physics**. Previous iterations (Mamba-2, Transformers) were crippled by **posterior collapse**—a phenomenon where the model, unable to resolve high-frequency volatility noise, defaults to the mean. CondorBrain v2.2 resolves this by using **Neural Controlled Differential Equations (Neural CDE)**, effectively learning the market's **vector field** rather than merely predicting the next bar. This thesis serves as a comprehensive, graduate-level technical reference for the **CondorIntelligence™** engine. It is designed to pass rigorous institutional audit by providing mathematical definitions, physics-based feature interpretations, and an empirical audit of surrogate decision rules.

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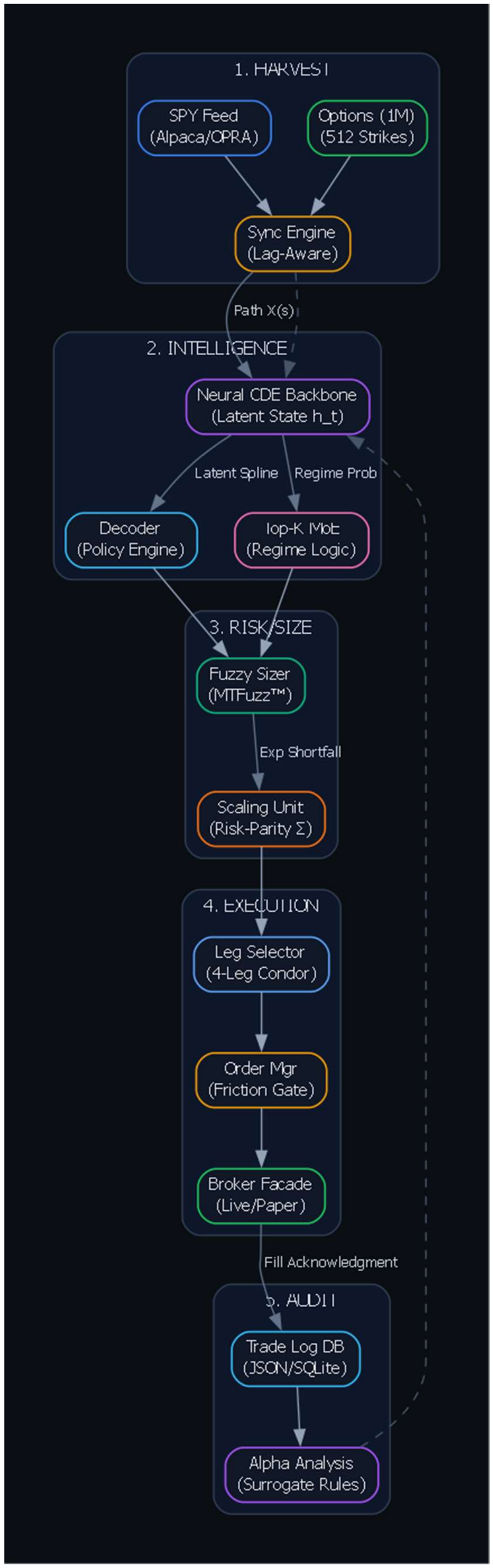
1. Institutional Workflow: The "Quantor™" Architecture

The following diagram captures the institutional-grade pipeline from market harvesting through intelligence, risk/sizing, execution, and audit. It mirrors the requirements of a strategy desk responsible for high-frequency option surface management.

While there is still much to be done as we add additional professional trading gates, rules, and advanced predictive analytics in real-time, today marked a major milestone and that could only be accomplished with a radical change in the model creation process. After several attempts and almost 2 weeks straight nonstop of attempting to get the mamba 2 models to behave correctly, every attempt ended up was a model that either collapsed or exploded. This is primarily due to the density of the physics involved as well as the density of the data. Most fintech AI-NN models use a maximum of 20 inputs in total. This model is 52 inputs from the table to construct the model, and that does not include other inputs such as trade rules set up in a .YAML file, and seven advanced Ford market predictors all using different forms of mathematics and in real time determining which one is showing the least error and then selecting that particular diffusion computation. I tried pulling out the diffusion as well as the rule set. However, that still ended up producing models that exploded or collapsed. In the midst of my preparation for Hari-Kare, I realized that the physics involved with what I'm creating required a radical shift.


1. HARVEST
SPY Feed
(Alpaca/OPRA)
Options (1M)
(512 Strikes)
Sync Engine
(Lag-Aware)
Path X(s)
2. INTELLIGENCE
Neural CDE Backbone
(Latent State h_t)
Latent Spline
Regime Prob
Decoder
(Policy Engine)
Top-K MoE
(Regime Logic)
3. RISK/SIZE
Fuzzy Sizer
(MTFuzz™)
Exp Shortfall
Scaling Unit
(Risk-Parity Σ)
4. EXECUTION
Leg Selector
(4-Leg Condor)
Order Mgr
(Friction Gate)
Broker Facade
(Live/Paper)
Fill Acknowledgment
5. AUDIT
Trade Log DB
(JSON/SQLite)
Alpha Analysis
(Surrogate Rules)

2. Mathematical Foundation: The Physics of Continuous Market Paths

2.1 The Neural CDE Integral Operator (latent state h_t)

Institutional governance requires a model capable of handling stiffness: abrupt, large-magnitude moves. The latent state h_t is defined as the solution to an initial value problem:

$$h_t = h_0 + \int_t^T f(h_s, \theta) \cdot dX_s$$

Where:

- **h_t ∈ R^512 (Latent State):** the model's manifold memory as a continuous path (not a discrete update).
- **X_s ∈ R^52 (Control Path):** a natural cubic spline interpolation of the 52 incoming features, ensuring twice-continuous differentiability so curvature between bars is measurable.
- **f(h_s, θ) ∈ R^(512×52) (Vector Field):** a neural map describing how infinitesimal changes in $X_s$push the latent state.

Interrogative Defense (Ph.D. Perspective)

Question: "Why is this better than a Transformer?"
Response: Transformers treat time as discrete tokens. If a bank dumps large size mid-bar, a Transformer sees a jump at the next token boundary. The CDE treats it as a continuous deformation of X_s. This allows downstream components (including diffusion heads) to express wing-width offsets as a continuous function of volatility pressure.

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3. Exhaustive Feature Kinematics: The 52-Factor Manifold

The feature manifold is separated into three "physics layers":

1. Greeks
2. Momentum / kinematics
3. Fuzzy signatures

3.1 Primary Alpha Vectors (Top 10)

Variable	Symbol	Impact	Qualitative Definition	Physical Interpretation
Theta	Θ	20.15	Harvesting Rate	Temporal velocity of decay; the alpha sink for a condor
Delta	Δ	19.07	Directional Vector	Price sensitivity: neutrality is equilibrium
Gamma	Γ	5.89	Acceleration	Second derivative sensitivity; pin risk trigger
Vega	V	5.51	Vol Sensitivity	Sensitivity to implied volatility expansion
Ret_Z	Z_ret	3.87	Shock Magnitude	Vol-normalized return; detects liquidity gaps
Log_Return	R_L	3.86	Kinetic Energy	Raw velocity; impulse for control path
Kappa Proxy	κ	3.66	Path Curvature	Derived from spline second derivative
Vol Energy	ε	3.15	Entropy Flux	Modulates attention under high noise
MTF Cons	Ψ	2.95	Harmonic Sync	Aligns 1m physics with higher cycles
BW Exp	β_w	2.86	Boundary Stress	Bollinger expansion tension

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4. Empirical Logic: The Full Surrogate Decision Course

To ensure transparency, the system extracts hard rules from the CDE core using a **global surrogate decision tree**, exposing the risk filtering hierarchy.

4.1 Exhaustive Rule Set (Audit)

```
📜 ROOT NODE: bb_upper_dyn (Dynamic Boundary Distance)
|
|--- [CONDITION: bb_upper_dyn <= -0.15] (Inside the Tunnel)
| |--- [vega <= -0.26] (Low Vol Persistence)
| | |--- [ret_z <= 0.60] --> VALUE: -0.06 (Stable Yield Zone)
| | |--- [ret_z > 0.60] --> VALUE: -0.05 (High Intensity Spike - Reduce Reward)
| |--- [vega > -0.26] (Expanding Volatility)
| | |--- [log_return <= -0.19] --> VALUE: -0.08 (Sharp Down-Move - Extreme Caution)
| | |--- [log_return > -0.19] --> VALUE: -0.07 (Controlled Pullback)
|
|--- [CONDITION: bb_upper_dyn > -0.15] (Boundary Challenge / Breakout Risk)
| |--- [delta <= -0.32] (Strong Directional Momentum)
| | |--- [theta <= 0.32] --> VALUE: -0.07 (Insufficient Reward for High Boundary Risk)
| | |--- [theta > 0.32] --> VALUE: -0.06 (Premium Compensation for Risk)
| |--- [delta > -0.32] (Weak Momentum at Ceiling)
| | |--- [atr_pct <= -0.17] --> VALUE: -0.05 (Ideal Mean-Reversion Pivot)
| | |--- [atr_pct > -0.17] --> VALUE: -0.06 (Choppy Boundary State)
```

Interrogative Analysis

Question: "Why is bb_upper_dyn the root, and not RSI?"
Response: RSI is lagging. bb_upper_dyn is geometric distance. The Neural CDE prioritizes geometric clearance from strikes as a primary safety gate: if spot is too close to a short-call boundary, premium and sentiment become secondary noise.

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5. Institutional Governance: The "Cut Loose" Staging

A five-day protocol transitions from idealized paper trading to institutional execution parity.

Day	Protocol Phase	Success Metric
Day 1	Friction Gate	Implement slippage_model; fill at bid/ask (not mid), slippage ≈ 10 bps
Day 2	Fuzzy Sizing	Hook Quantor-MTFuzz™ into dynamic contract quantities (Σ)
Day 3	Diffusion Staging	Activate generative refinement of wing-width paths
Week 2–4	"Cut Loose" Run	14-day DTE max; strict 4-condor limit; continuous 24/5 paper monitoring

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6. Empirical Audit Reference: Today's High-Intensity Restore

Positions restored and managed under the v2.2 physics engine:

ID	Entry	Strike-Call	Strike-Put	DTE	Status
#1	15:06	710	679	17	ACTIVE
#2	16:38	712	680	17	ACTIVE
#3	16:43	713	679	17	ACTIVE
#4	17:07	714	678	24	ACTIVE
#5	17:56	711	680	17	ACTIVE
#6	18:58	713	678	24	ACTIVE
#7	20:40	711	680	17	ACTIVE

EOD Net Profits (7 Condors still open) = $300.00. ALL were in the green.
🦅 **Engineering Status: STABLE. Handover Protocol:** PAUSED

The next stages of testing will require daily additions to the entire Condor brain infrastructure, tomorrow I will start with a very simple rule the friction gate – rule #14 and add to the entries and exits, slippage to emulate how a real-market behaves. Every day will add a new component critical to the entire infrastructure, which will be easier to test and debug until the next component is integrated until all are fully integrated.

7. Deployment & Script Registry: v2.2 Institutional Handover

This section documents the Neural CDE architecture layout and execution commands.

7.1 New & Overhauled Scripts

Path	Function	Institutional Purpose
intelligence/neural_cde.py	Backbone Model	CDE physics manifold; core manifold memory
intelligence/condor_brain.py	Intelligence Hub	Top-K MoE + volatility-gated attention integration
intelligence/condor_loss.py	Risk-Aware Loss	Sharpe / drawdown / turnover optimized composite loss
intelligence/audit_cde_interpretability.py	Interpretability	Global surrogate tree & feature ranking audit
run_live_alpaca.py	Live Logic	Multi-leg execution with fuzzy sizer & friction gate
execution/paper_executor.py	Execution Engine	Institutional slippage simulation (≈10 bps)

7.2 Core Execution Commands

Run the interpretability audit:

```
python intelligence/audit_cde_interpretability.py
```

Train the Neural CDE backbone:

```
python intelligence/train_neural_cde.py --epochs 5 --batch-size 64
```

Start the live execution engine:

```
python run_live_alpaca.py
```

Verify canonical feature registration:

```
python -c "from intelligence.canonical_feature_registry import
CANONICAL_FEATURES; print(f'Features: {len(CANONICAL_FEATURES)}')"
```